Exercise 9

①

a)

We want to compute the gradient

$\nabla_\phi \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]$

The problem is that the distribution q we sample from itself depends on ϕ.

So with a change of ϕ, not only the integrand $\log p_\theta(x|z)$ changes but also where z is sampled from.

→ Write the gradient explicitly

$\mathbb{E}_{q_\phi(z|x)} [f(z)] = \int q_\phi(z|x) f(z) \, dz$

$\nabla_\phi \mathbb{E}_{q_\phi} [f(z)] \neq \mathbb{E}_{q_\phi} [\nabla_\phi f(z)]$

The correct gradient has two terms:

$\nabla_\phi \mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)]$

$= \nabla_\phi \int q_\phi(z|x) \log p_\theta(x|z) \, dz$

$= \int q_\phi(z) \nabla_\phi \log q_\phi(z) \log p_\theta(x|z) \, dz$

$= \mathbb{E}_{q_\phi} [\log p_\theta(x|z) \nabla_\phi \log q_\phi(z|x)]$

This is problematic:

- Monte Carlo estimates of this term tend to have high variance
  ↳ Training is slow and unstable
- No direct backpropogation through the sampling is possible
  $\nabla_\phi \log q_\phi$ forces differentiation throug the sampling distribution itself

→ gradient based optimization very difficult.

b)

$\epsilon \sim \mathcal{N}(0, I)$ defines the transformation $z = \mu_\phi(x) + \sigma_\phi(x) \odot \epsilon = g_\phi(x, \epsilon)$

Then

$\mathbb{E}_{q_\phi(z|x)} [\log p_\theta(x|z)] = \int q_\phi(z|x) \log p_\theta(x|z) \, dz$

$z = g_\phi(x, \epsilon)$ is a bijection from $\epsilon \in \mathbb{R}^k$ to $z \in \mathbb{R}^k$

so $q_\phi(z|x)\,dz = p(\epsilon)\,d\epsilon$ with $p(\epsilon) \sim \mathcal{N}(0, I)$

$$= \int \log p_\theta(x | g_\phi(x, \epsilon))\, p(\epsilon)\, d\epsilon$$

$$= \mathbb{E}_{\epsilon \sim \mathcal{N}(0,I)} \left[\log p_\theta(x | g_\phi(x, \epsilon)) \right]$$

$$= \quad '' \quad \left[\log p_\theta(x | z(\epsilon, \phi)) \right]$$

Before, z was sampled from a ϕ-dependent distribution

Now, $z = g_\phi(\epsilon)$ with ϵ a fixed distribution, gradient can be calculated with the standard chain rule.

-> every dependence on ϕ is inside a deterministic map in g_ϕ

c)

We have

$$q_\phi(z_{1:T} | x_{1:T}) = \prod_{t=1}^{T} q_\phi(z_t | x_{\leq t})$$

This means each z_t is conditionally independent given the past observations $x_{\leq t}$.

(Not necessarily independent of earlier $z_{<t}$)

The reparametrization at looks like:

$$z_t = \mu_\phi(x_{\leq t}) + \sigma_\phi(x_{\leq t}) \odot \epsilon_t \quad , \quad \epsilon_t \sim \mathcal{N}(0, I)$$

-> allows reparametrization per time step

We also have

$$p(z_1) = \mathcal{N}(0, I) \,, \quad p(z_t | z_{t-1}) = \mathcal{N}(A z_{t-1}, \Sigma)$$

(introduces time dependence across time)

The KL-term in the ELBO then is:

$$D_{KL}(q_\phi(z_{1:T} | x_{1:T}) \| p(z_{1:T}))$$

-> couples latent variables across time

=> Reparametrization per time step possible, but not time independent

(temporal structure affects gradient computation, not the reparametrization)